Cover Letter

Dear Editor,

I am pleased to submit my manuscript entitled "Public HBV genotype B/C core diversity reveals an HLA-DQB1*03:01-associated class-II presentation gap" for consideration as an Article in Viruses.

HLA-DQB1*03:01 has recently been associated with increased risk of HBV-related hepatocellular carcinoma, and prior work suggested reduced predicted binding of HBV nucleocapsid peptides by DQB1*03:01-containing HLA-DQ molecules. However, it has remained unclear whether this finding reflects a reference-sequence observation or a presentation gap that persists across naturally occurring HBV genotype B/C sequence diversity.

In this public-data computational study, I analyzed 1576 quality-controlled GenBank HBV genotype B/C complete-genome records, generated 11176 unique HBV core 15-mer peptide windows, and performed HLA-DQ heterodimer MHC-II binding prediction using the IEDB NetMHCIIpan workflow. I found that two DQB1*03:01-containing heterodimers were marked low-binding outliers in an expanded seven-heterodimer HLA-DQ panel, with occurrence-weighted core binder rates of 0.23%-0.80% compared with 7.66%-13.52% for non-risk comparison heterodimers. The gap persisted after method sensitivity analysis, exact and identity-based sequence de-redundancy, record-level bootstrap, genotype/region/year stratification, IEDB human T-cell epitope overlap analysis, and reference-proteome controls.

The manuscript is positioned as a hypothesis-generating public-data immunogenomics study. I do not claim experimental proof of antigen processing, surface presentation, CD4 T-cell recognition, or HBV-HCC causality. Instead, the study provides a reproducible computational extension of the DQB1*03:01-HBV-HCC nucleocapsid axis into public viral sequence diversity and nominates HBV core antigen presentation as a focused target for future validation.

All data used in this study are public. The analysis scripts, processed tables, figures, and one-step reproduction instructions are available in a public GitHub repository at https://github.com/llf48/HBV-DQB10301-public-immunogenomics, with a Zenodo archival release at https://doi.org/10.5281/zenodo.19956882. No individual-level human participant data were accessed.

I confirm that this manuscript is original, has not been published previously, and is not under consideration elsewhere. As the sole author, I have approved the submitted version and agree with its submission to Viruses. I declare no competing interests and no external funding.

I am a student author at The First School of Clinical Medicine, Southern Medical University, and this independent public-data study has no grant, laboratory, or principal-investigator APC support. I have therefore prepared a separate request for a full article processing charge waiver for the Editorial Office's consideration.

Thank you for considering my work.

Sincerely,

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